KLAi

Matrices: tables of numbers that also move vectors, and the heart of every AI layer

A language model has billions of parameters. What are those numbers, and what does the model do with them?

Mathematics for AI and Machine Learning | Session 3 of 18 | Linear algebra

KLAi Professional Learning | Jotham Martin Wambi

Today in one view: what you will leave with

By the end of today you can



Read a matrix as a table and as a transformation, and state its shape (rows by columns)

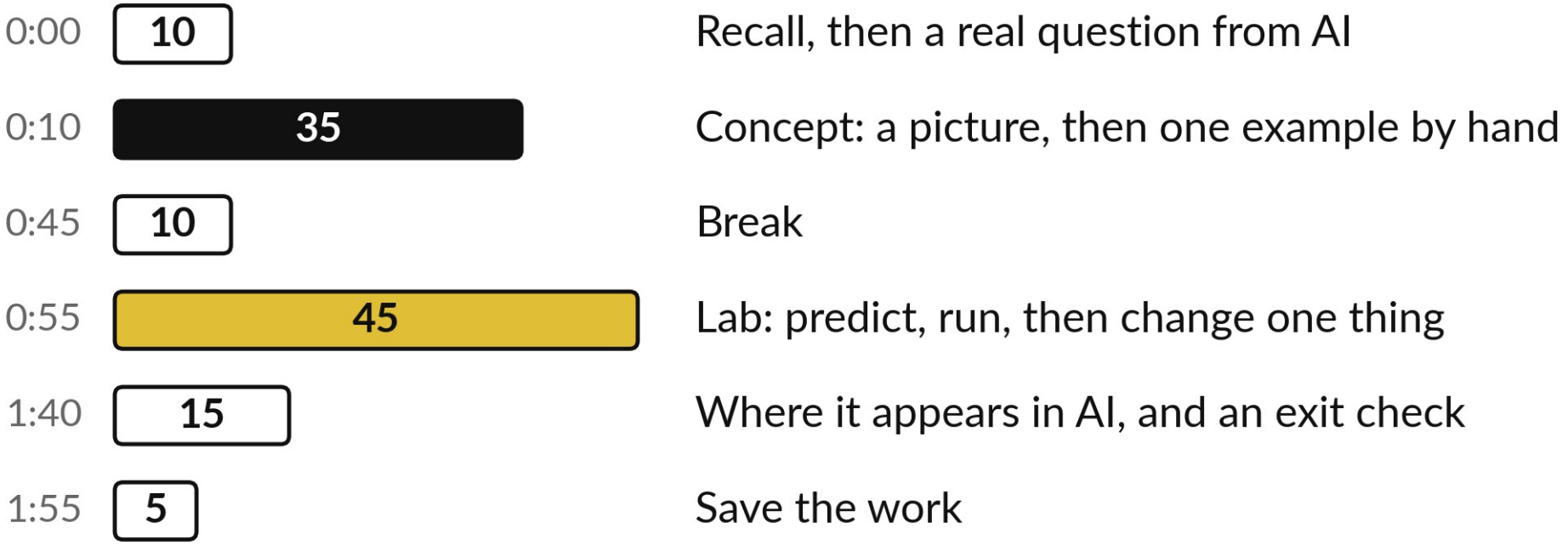


Multiply a matrix by a vector and a matrix by a matrix by hand, and say when the shapes allow it



Compute a layer y = W x + b, and count the parameters of a small network, in Python and in a spreadsheet

Your two hours: matrices, then the lab, then parameters



Pictures and hand calculations first. Then the lab on products, layers and parameters.

Three questions from last time

1. What is the dot product of (1, 2, 3) and (4, 0, 2)? Is the result a vector or a number?

2. What is the L2 norm (length) of the vector (2, 3, 6)?

3. Why does a search tool rank by cosine similarity and not by the raw dot product?



Answer aloud first. Then we check together.

Answers to the three questions from Session 2

Question	Short answer
What is the dot product of (1, 2, 3) and (4, 0, 2)? Is the result a vector or a number?	10. It is one number: $1 \times 4 + 2 \times 0 + 3 \times 2$.
What is the L2 norm (length) of the vector (2, 3, 6)?	7, because 4 + 9 + 36 = 49 and the square root of 49 is 7.
Why does a search tool rank by cosine similarity and not by the raw dot product?	Cosine similarity divides by both lengths, so a long vector does not win just for being long. Only the direction counts.

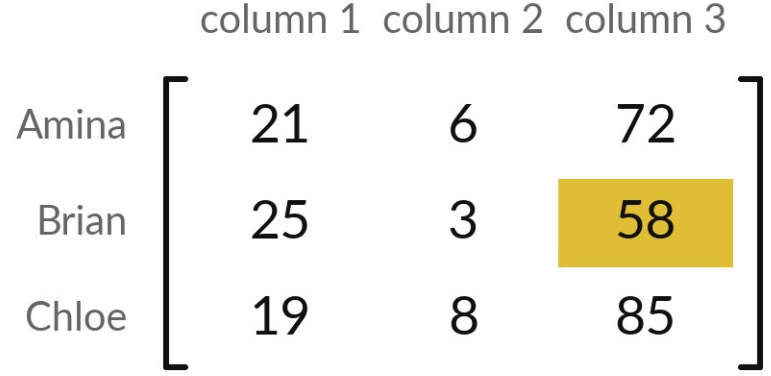
If one answer surprised you, look again at the Session 2 slides before the lab.

Words we use today

	Word	What it means
	**Matrix**	A table of numbers with m rows and n columns. Its shape is m × n.
	**Matrix multiplication**	A rule that combines two matrices. Each entry of the answer is a dot product of a row and a column.
	**Transpose**	The matrix flipped so that rows become columns. It is written with a small T.
	**Identity matrix**	A square matrix of 1s on the diagonal and 0s elsewhere. It leaves every vector unchanged.
	**Parameters**	The numbers inside a model that training sets: its weights W and biases b.

A matrix is a table of numbers with rows and columns

A is a 3 × 3 matrix: 3 rows and 3 columns



Matrix: a table of numbers. Here, three learners from Session 2, one per row: age, hours, score.

Each row is a vector: Amina = (21, 6, 72).

Gold entry: row 2, column 3 is 58, Brian's score.

Illustrative (made-up numbers to show the arithmetic).

Matrix notation: a capital letter, a shape and an entry written with two small numbers

Shape $A \in \mathbb{R}^{m \times n}$

Entry in row i, column j A_{ij}

Our table $A \in \mathbb{R}^{3 \times 3}, \qquad A_{23} = 58$

Symbol	What it means	In our data
A	A matrix is written with a capital letter. A vector is a lowercase bold letter.	A = the learner table
m × n	The shape: m rows by n columns. Rows are always named first.	3 × 3
First formula	Reads: A is a table of ordinary (real) numbers with m rows and n columns.	A has 3 rows and 3 columns
Aij	The entry in row i, column j of A (small i and j below the A).	A_{23} = 58

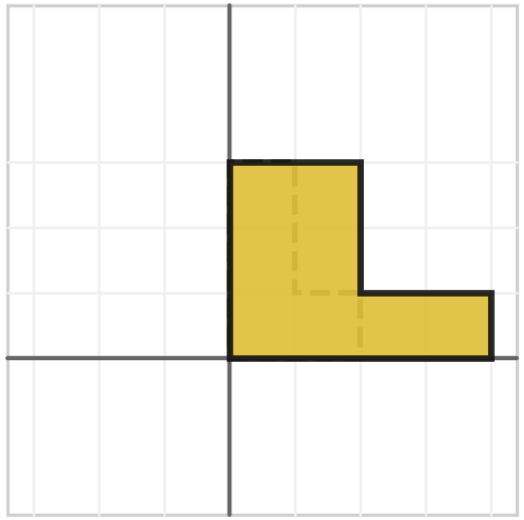
Python counts from 0, so A_{23} has row index 1 and column index 2. A spreadsheet counts from 1: INDEX(range, 2, 3).

A matrix can also move points: scaling, rotating and shearing a shape

Scale

rows (2, 0) and (0, 1)

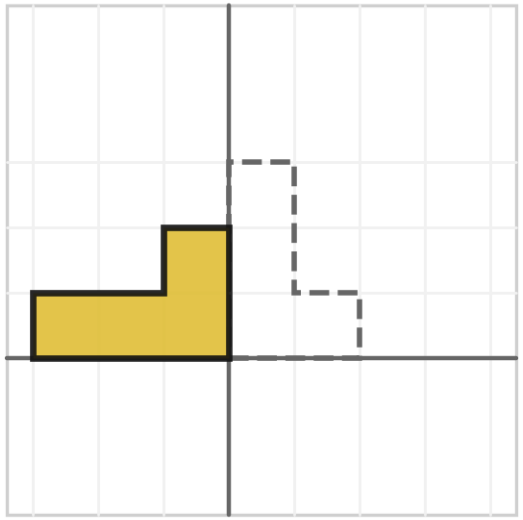
doubles every x, keeps y



Rotate

rows (0, −1) and (1, 0)

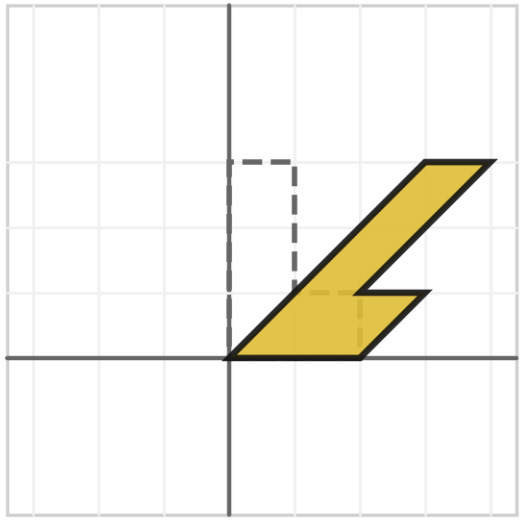
quarter turn anticlockwise



Shear

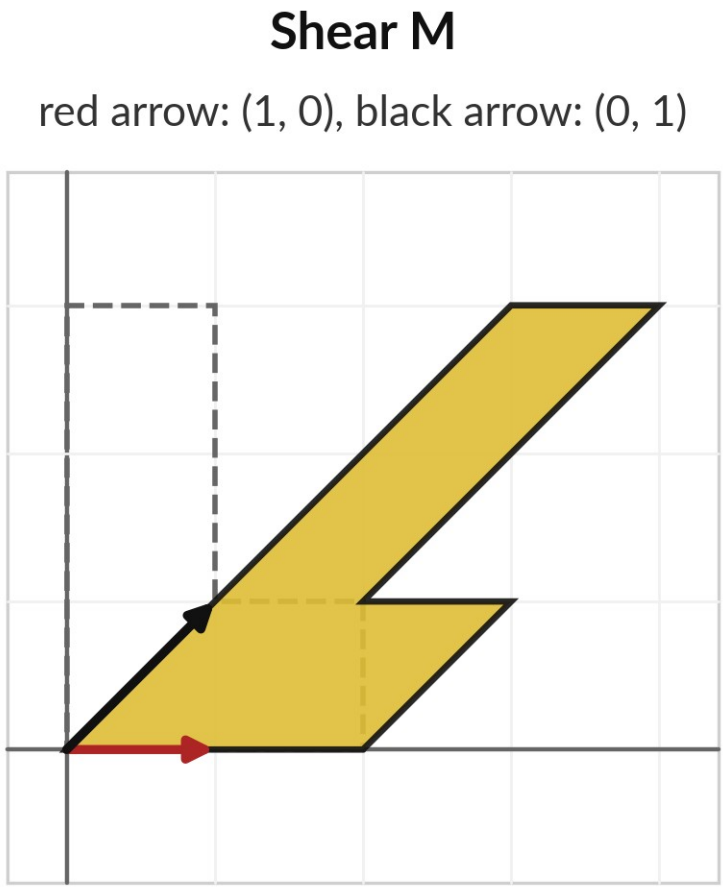
rows (1, 1) and (0, 1)

slides each point right by its height



Grey dashed: the original shape. Gold: the result. The rotation sends the corner (1, 3) to (−3, 1).

The columns of a matrix show where the two axis arrows land



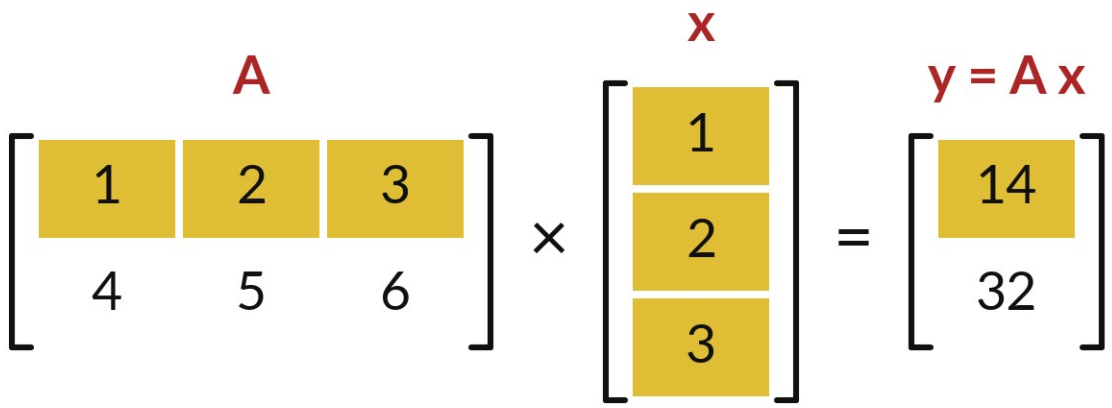
M has rows (1, 1) and (0, 1). Red arrow (1, 0) lands on (1, 0). Black arrow (0, 1) lands on (1, 1). These two are the columns of M.

Every other point follows: the grid tilts.

Corner (1, 3) lands on (4, 3). The next two slides show the rule and the working.

To read a transformation, read the columns.

Matrix times vector: each entry of the answer is a dot product of one row with the vector



The gold row of A, times x, gives the gold entry of y

$$1 \times 1 + 2 \times 2 + 3 \times 3 = 14$$

Multiply A (2 × 3) by x (3 entries). Dot each row of A with x in turn.

Row 1: 1 × 1 + 2 × 2 + 3 × 3 = 14, the gold entry.

Row 2: 4 × 1 + 5 × 2 + 6 × 3 = 32.

The answer y = (14, 32) has one entry per row of A.

Matrix times vector as a formula, and the same calculation for a point of the shape

Matrix times vector

$$y_i = \sum_{j=1}^{n} A_{ij} x_j$$

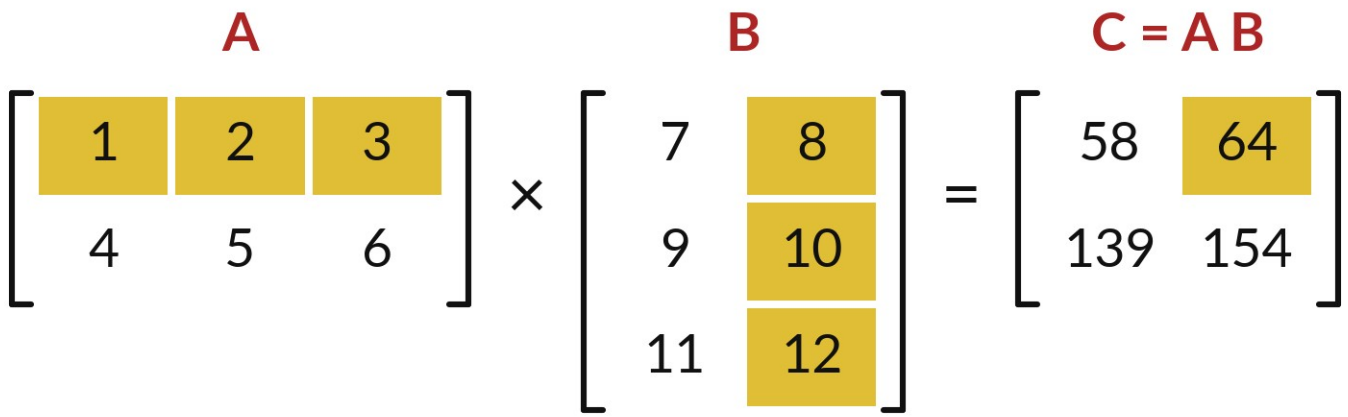
Shear, corner (1, 3)

$$1 \cdot 1 + 1 \cdot 3 = 4, \qquad 0 \cdot 1 + 1 \cdot 3 = 3$$

Symbol	What it means	In our data
Aij	Entry in row i, column j of A.	$A_{12} = 2$
xj	Entry j of the vector x. It meets column j of A.	$x_2 = 2$
yi	Entry i of the answer: the dot product of row i of A with x.	$y_1 = 14$
Σ, j, n	Add up over j = 1 to n, the columns of A.	n = 3
Shear on (1, 3)	Row 1 of M with (1, 3), then row 2 of M with (1, 3).	The point (1, 3) lands on (4, 3)

Python: A @ x. Spreadsheet: =MMULT(A, x) as an array formula.

Matrix times matrix: each entry of the answer is a dot product of a row of A with a column of B



Gold row of A, gold column of B: together they give the gold entry of C

$$1 \times 8 + 2 \times 10 + 3 \times 12 = 64$$

Multiply A (2 × 3) by B (3 × 2). Entry (i, k) of the answer is row i of A dotted with column k of B.

Gold: row 1 of A, column 2 of B: 8 + 20 + 36 = 64.

C = A B has rows (58, 64) and (139, 154).

C has the rows of A and the columns of B: 2 × 2.

The matrix product as a formula, with a count of the multiplications

Matrix product

$$C_{ik} = \sum_{j=1}^{n} A_{ij} B_{jk}$$

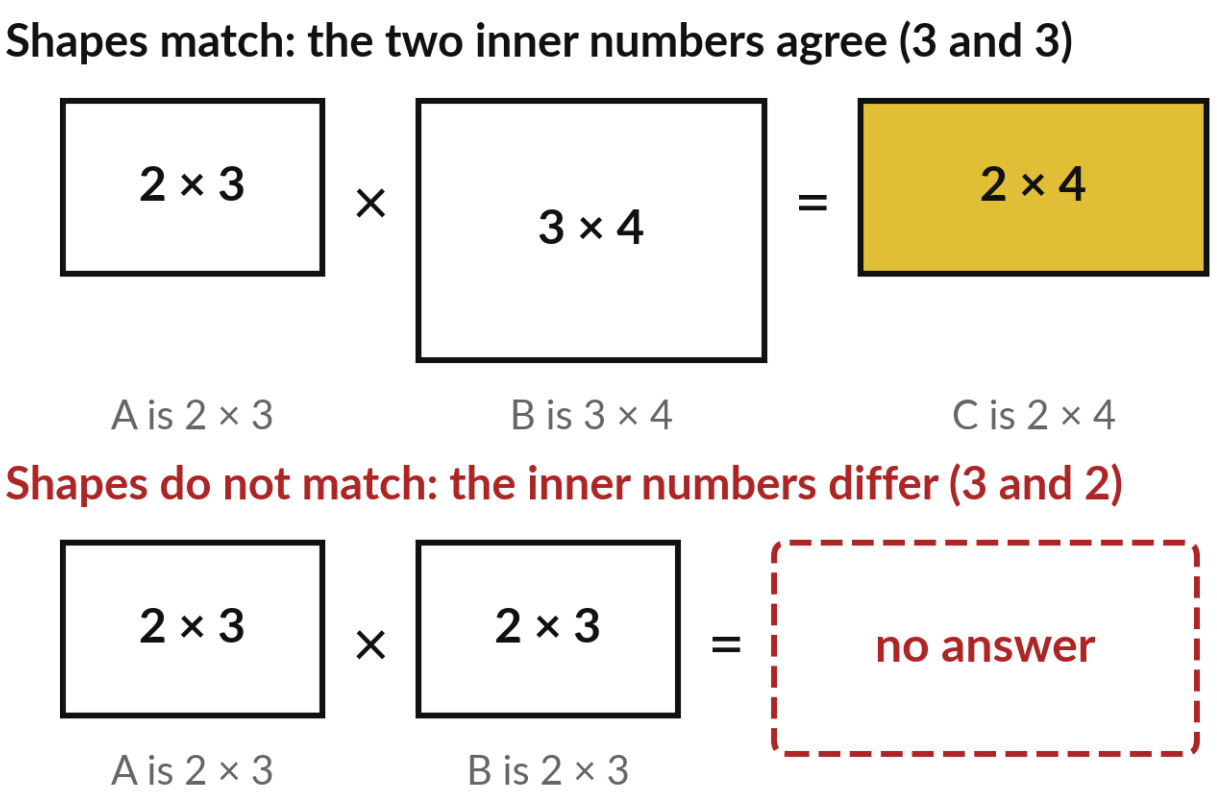
Our entry

$$C_{12} = 1 \cdot 8 + 2 \cdot 10 + 3 \cdot 12 = 64$$

Symbol	What it means	In our data
A, B, C	A is m × n, B is n × p. Then C = A B is m × p.	C is 2 × 2
Σ, j, n	Add up over j = 1 to n.	j = 1, 2, 3
Cik	Row i of A dotted with column k of B.	C_{12} = 64
Cost	m × p entries, n multiplications each.	2 × 3 × 2 = 12

Python: A @ B. Spreadsheet: =MMULT(A, B) over a 2 × 2 range, entered as an array formula.

The shapes must match: the two inner numbers agree, and the outer numbers are the shape of the answer



A (m × n) times B (n × p) works when the columns of A equal the rows of B. The answer is m × p.

2 × 3 times 3 × 4 works: inner 3 and 3. The answer is 2 × 4.

2 × 3 times 2 × 3 fails: a row has 3 entries, a column 2.

Check the shapes first.

The order matters: A B and B A are different matrices

A B

A 2 × 3, B 3 × 2.

A B is 2 × 2, rows (58, 64) and (139, 154).

Works: inner 3 and 3.

B A

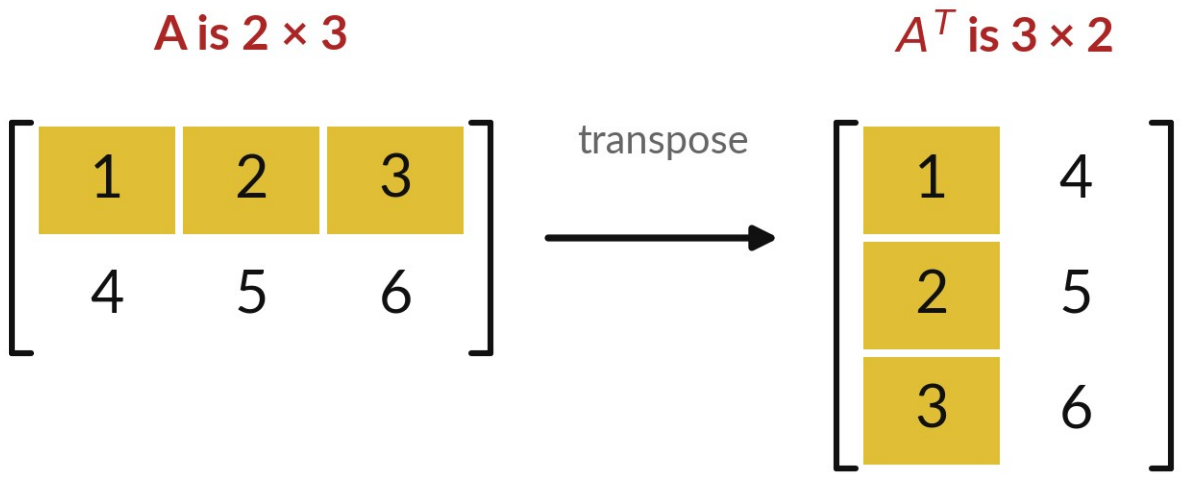
B 3 × 2, A 2 × 3.

B A is 3 × 3, first row (39, 54, 69).

Works: inner 2 and 2.

The order matters: A B is usually different from B A.

The transpose flips rows into columns



Rows become columns. Entry (i, j) of A is entry (j, i) of the transpose.

Transpose: write each row of A as a column. The result is written A^T, with a small T.

A is 2 × 3, so A^T is 3 × 2.

Entry (i, j) of A becomes entry (j, i).

Dot product: $u \cdot v$ equals $u^T v$.

The dot product is a matrix product of a row and a column

Dot product as a matrix product

$$\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T \mathbf{v}$$

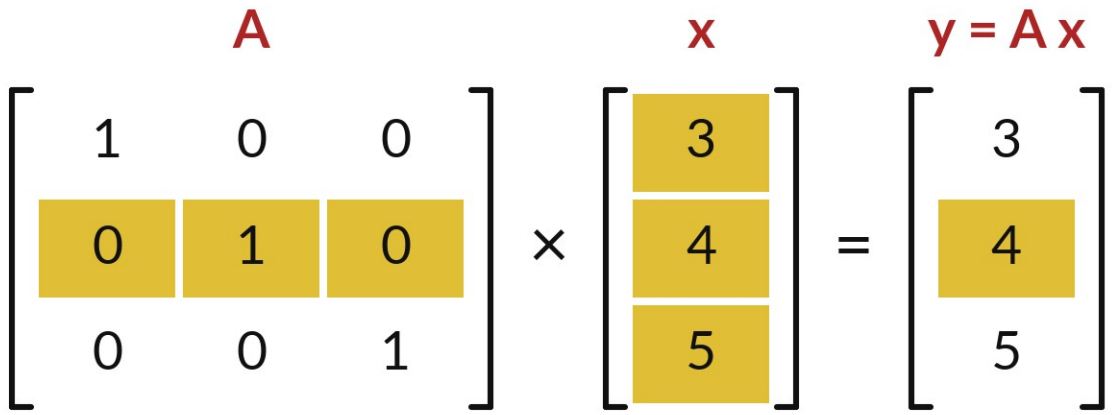
Our numbers

$$(1,\ 2,\ 3) \cdot (4,\ 0,\ 2) = 1 \cdot 4 + 2 \cdot 0 + 3 \cdot 2 = 10$$

Symbol	What it means	In our data
$\mathbf{u}^T$	The transpose of u: a row of 3 numbers, a 1 × 3 matrix.	(1, 2, 3)
$\mathbf{v}$	A column of 3 numbers, a 3 × 1 matrix.	(4, 0, 2)
$\mathbf{u}^T \mathbf{v}$	Shapes: 1 × 3 times 3 × 1. The inner numbers match and the answer is 1 × 1: one number.	10

By convention a vector is a column. So the dot product of Session 2 and today's matrix product are one idea.

The identity matrix leaves every vector unchanged



The gold row of A, times x, gives the gold entry of y

$$0 \times 3 + 1 \times 4 + 0 \times 5 = 4$$

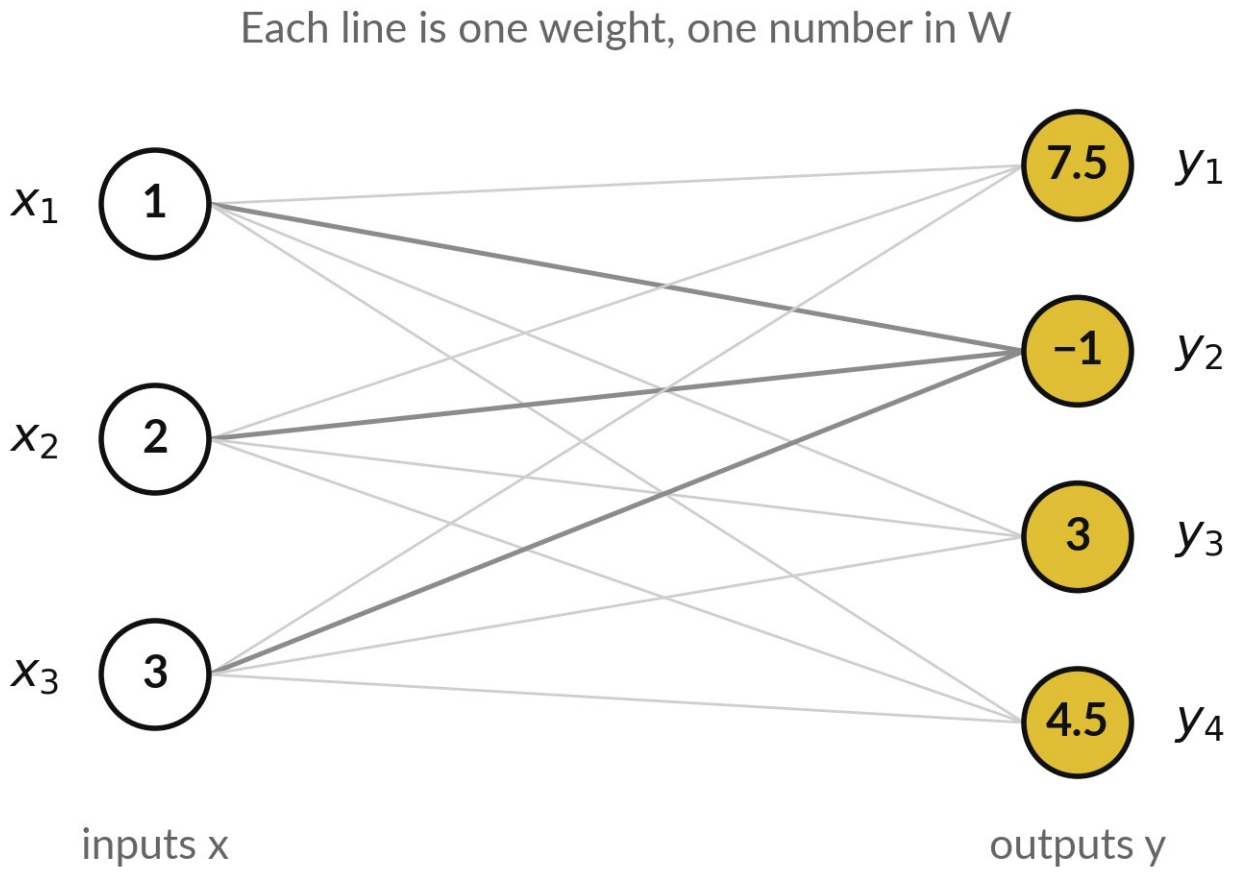
Identity matrix I: 1s on the diagonal, 0s everywhere else. For any vector x, I x = x.

Each row of I picks one entry of x. Row 2 picks 4.

It is the matrix version of multiplying by 1.

Session 4 uses it to define the inverse.

A linear layer computes y = W x + b; real layers then apply an activation function



A neural network is a chain of layers. This layer turns 3 inputs x into 4 outputs y. W is the weight matrix (4 × 3), b the bias vector.

Illustrative: x = (1, 2, 3).

Output 1: 1 × 1 + 0 × 2 + 2 × 3 = 7, plus bias 0.5 gives 7.5.

Training sets W and b: the parameters.

The layer y = W x + b, with every symbol named and worked in full

One linear layer

$$\mathbf{y} = W\mathbf{x} + \mathbf{b}$$

Our numbers

$$\mathbf{y} = (7.5,\ -1,\ 3,\ 4.5)$$

Symbol	What it means	In our data
x	The input vector, 3 entries.	(1, 2, 3)
W	The weight matrix, 4 × 3: 4 outputs, 3 inputs.	rows (1, 0, 2), (0, 1, −1), (2, 1, 0), (−1, 1, 1)
W x	Four dot products: 7, −1, 4 and 4.	(7, −1, 4, 4)
b	The bias vector. It is added entry by entry.	(0.5, 0, −1, 0.5)
y	The output: W x + b.	(7.5, −1, 3, 4.5)

A real layer then applies an activation function to each output (Session 11). Python: W @ x + b.

Layer 2 takes the 4 outputs of layer 1 as its inputs

Layer 2

$$\mathbf{y}_2 = W_2\,\mathbf{y}_1 + \mathbf{b}_2$$

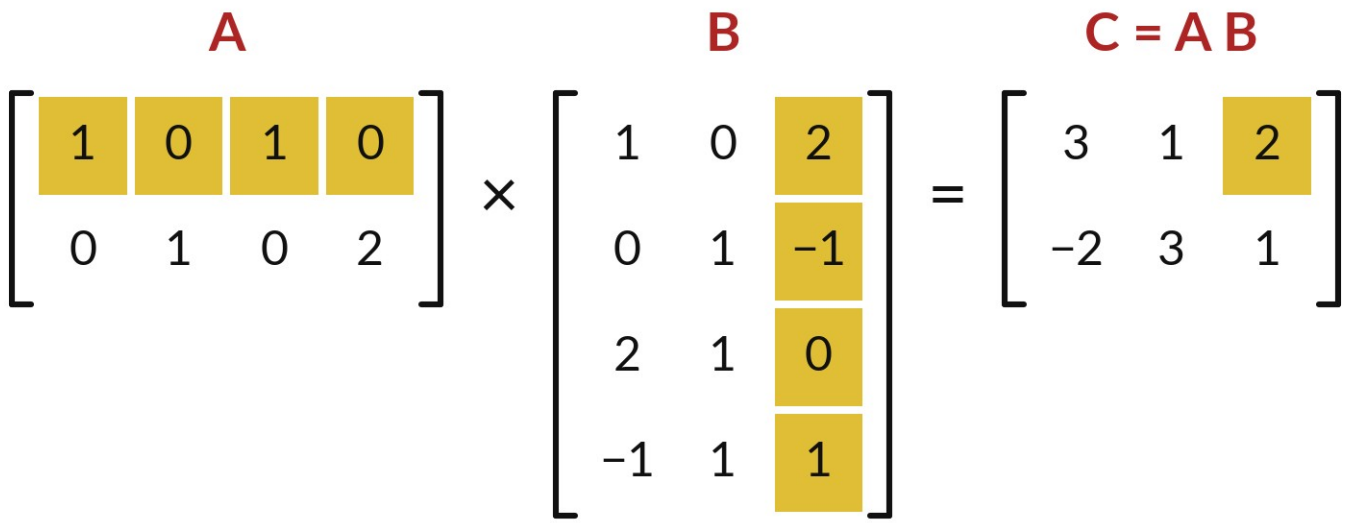
Our numbers

$$W_2\,\mathbf{y}_1 = (10.5,\ 8), \qquad \mathbf{y}_2 = (10.5,\ 9)$$

Symbol	What it means	In our data
$\mathbf{y}_1$	The output of layer 1, which is the input of layer 2.	(7.5, −1, 3, 4.5)
$\mathbf{W}_2$	The weight matrix of layer 2, 2 × 4.	rows (1, 0, 1, 0) and (0, 1, 0, 2)
$\mathbf{b}_2$	The bias vector of layer 2, 2 entries.	(0, 1)
$\mathbf{y}_2$	The output of layer 2: $W_2\ y_1 + b_2$.	(10.5, 9)

Activation function: a fixed nonlinear function applied to each output (Session 11). We leave it out today.

Stacking two layers is a matrix times a matrix



Gold row of A, gold column of B: together they give the gold entry of C

$$1 \times 2 + 0 \times (-1) + 1 \times 0 + 0 \times 1 = 2$$

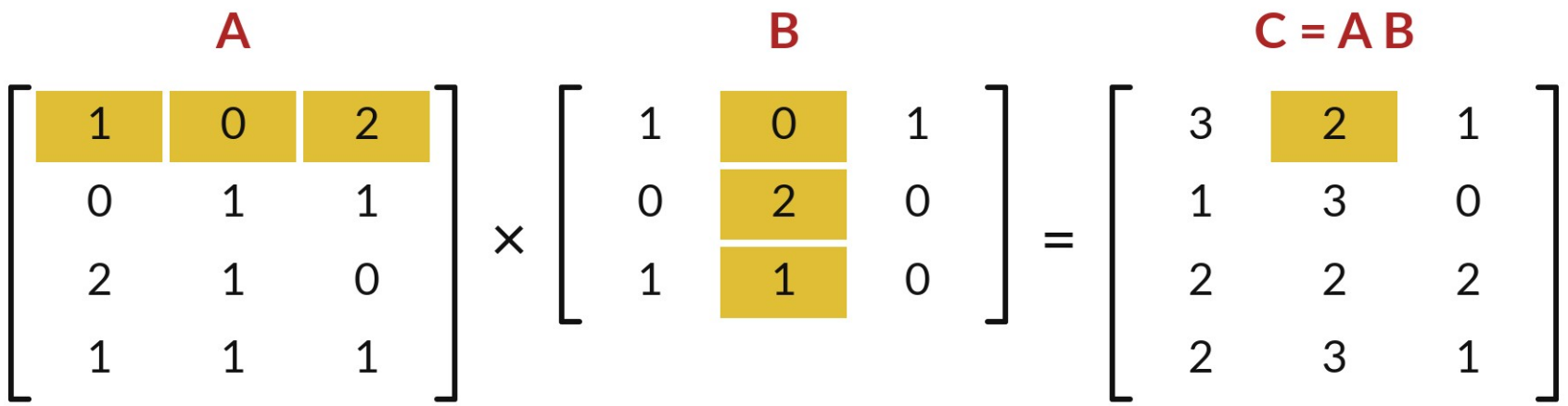
With no biases, layer 2 after layer 1 is W2 (W1 x) = (W2 W1) x. Multiply the weights first.

W2 W1 is 2 × 3: rows (3, 1, 2) and (−2, 3, 1).

On x = (1, 2, 3) it gives (11, 7), as two layers in turn do.

With no activation function, two layers collapse into one.

A whole sentence goes through a layer at once: stack its token vectors as the rows of a matrix

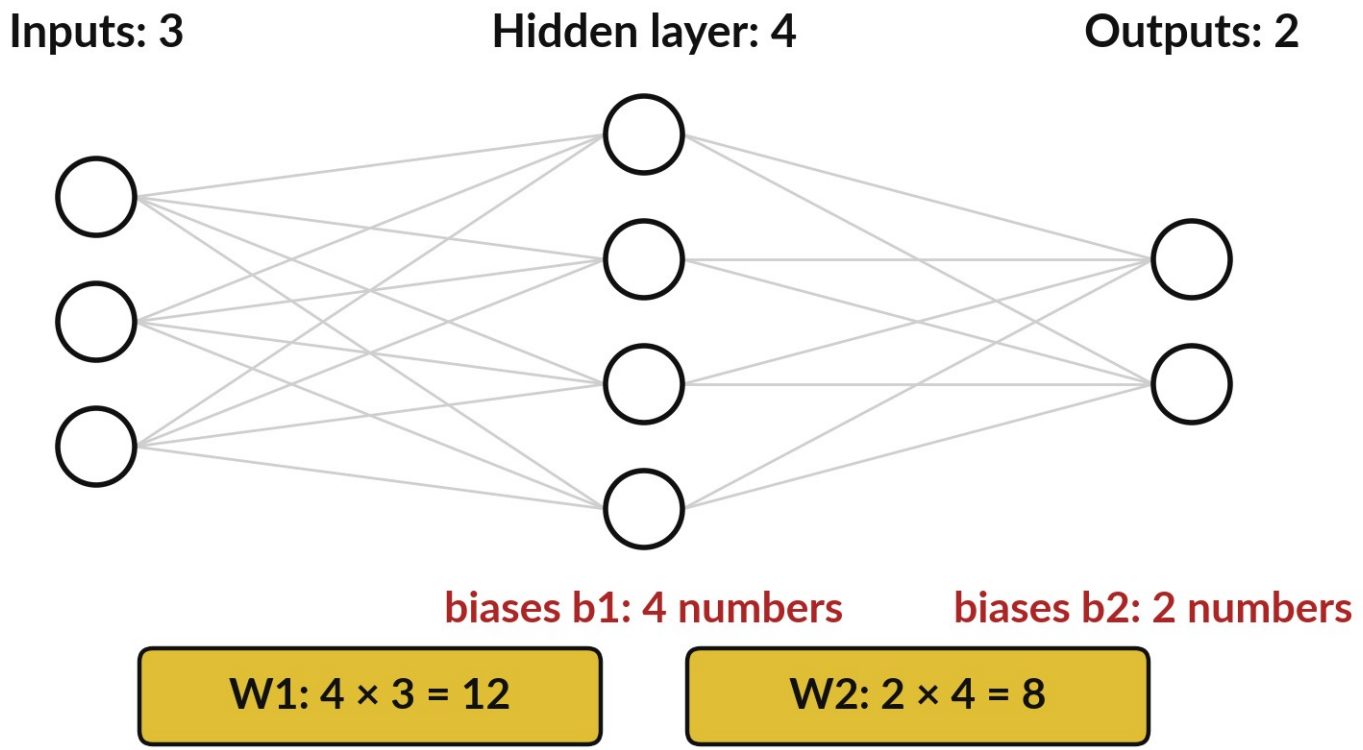


Gold row of A, gold column of B: together they give the gold entry of C

$$1 \times 0 + 0 \times 2 + 2 \times 1 = 2$$

Illustrative numbers. One token per row, so the product is X W, where one vector used W x.

Counting the numbers in the matrices gives the parameters of a small network



Parameters: the numbers a model learns. Toy network: 3 inputs, a hidden layer of 4, 2 outputs. A hidden layer sits between inputs and outputs.

Layer 1: W1 is 4 × 3 = 12 weights, plus 4 biases.

Layer 2: W2 is 2 × 4 = 8 weights, plus 2 biases.

Total: 12 + 4 + 8 + 2 = 26 parameters.

Parameters of a layer: m × n weights plus m biases

Parameters of one layer

$$\text{parameters} = m \times n + m$$

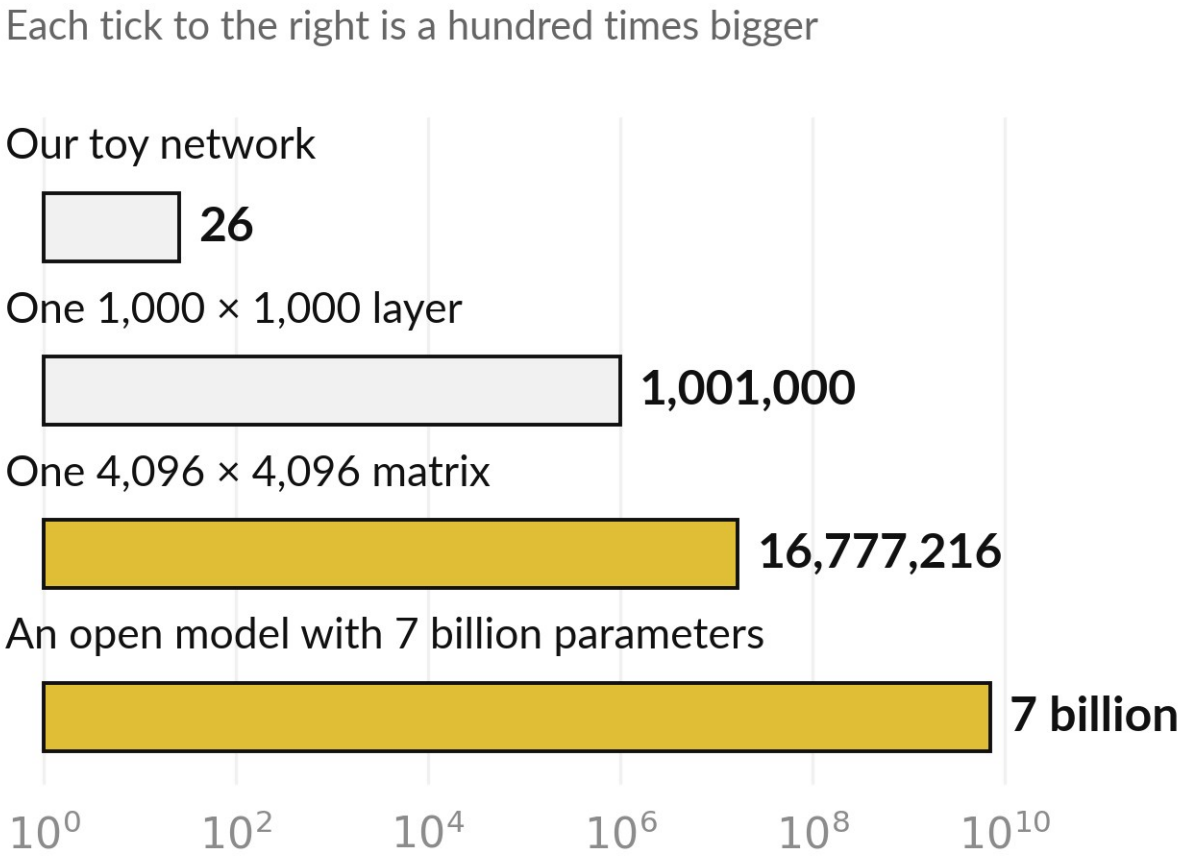
Our network

$$(4 \times 3 + 4) + (2 \times 4 + 2) = 16 + 10 = 26$$

Symbol	What it means	In our data
n	Number of inputs of the layer, which is the number of columns of W.	layer 1: n = 3
m	Number of outputs, which is the number of rows of W and the length of b.	layer 1: m = 4
m × n	The weights in W.	12
+ m	The biases in b.	4
Bigger	A layer from 1,000 inputs to 1,000 outputs.	1,000 × 1,000 + 1,000 = 1,001,000

Python: W.size + b.size. Spreadsheet: ROWS(W) * COLUMNS(W) + ROWS(b).

Billions of parameters means billions of numbers in matrices like these



Open models publish their size: Llama 2 has 7, 13 and 70 billion parameters (Touvron et al., 2023).

Logarithmic scale, as in Session 1.

At 2 bytes each, 7 billion numbers fill 14 gigabytes.

Training sets every one.

Common mistake: multiplying two matrices entry by entry instead of row by column

Wrong

Multiply matching entries of A and B.

Python: A * B, entry by entry. For our 2 × 3 and 3 × 2 it fails.

Right

Dot products of rows of A with columns of B.

Python: A @ B. Spreadsheet: MMULT.

Check the shapes first.

The star (*) multiplies entry by entry. The at sign (@) gives the matrix product.

Read the recipe before you run it: matrices in NumPy

```
A = np.array([[1, 2, 3], [4, 5, 6]])
x = np.array([1, 2, 3])
y = A @ x
C = A @ B
At = A.T
I = np.eye(3)
z = W @ x + b
n = W.size + b.size
```

In plain words

Line 1: a matrix from a list of rows. A.shape gives (2, 3).

Lines 3 and 4: the at sign is the matrix product. Spreadsheet: MMULT.

Line 5: the transpose. Spreadsheet: TRANSPOSE.

Line 6: the identity matrix of size 3 × 3.

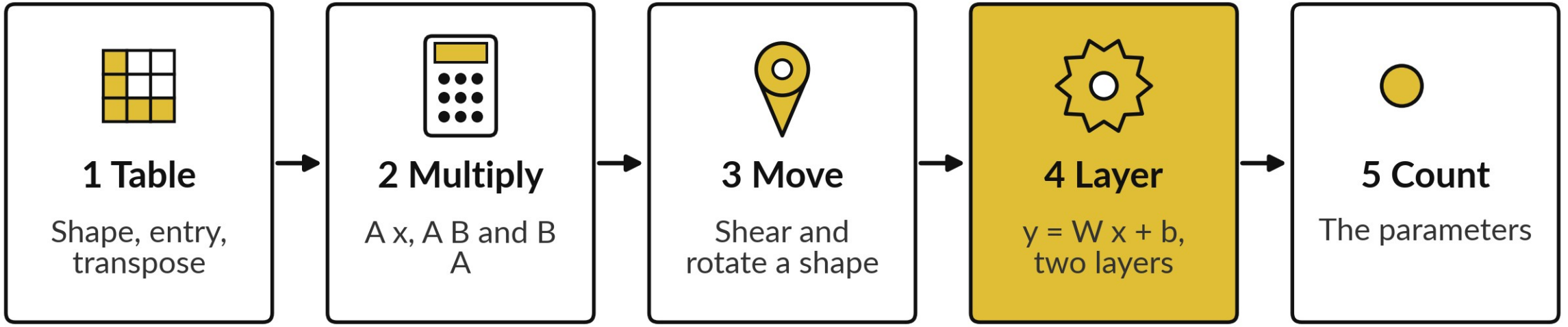
Line 7: one layer. Line 8: the number of parameters. Python counts from 0, so row 2 and column 3 is index 1, 2.



Part 2: the lab

Close the slides. Open the notebook or the workbook.

The lab has five steps, from a table to a count of parameters



Predict, run, read, change one thing. Workbook sheets: Products and Layers.

Check your numbers before you move on

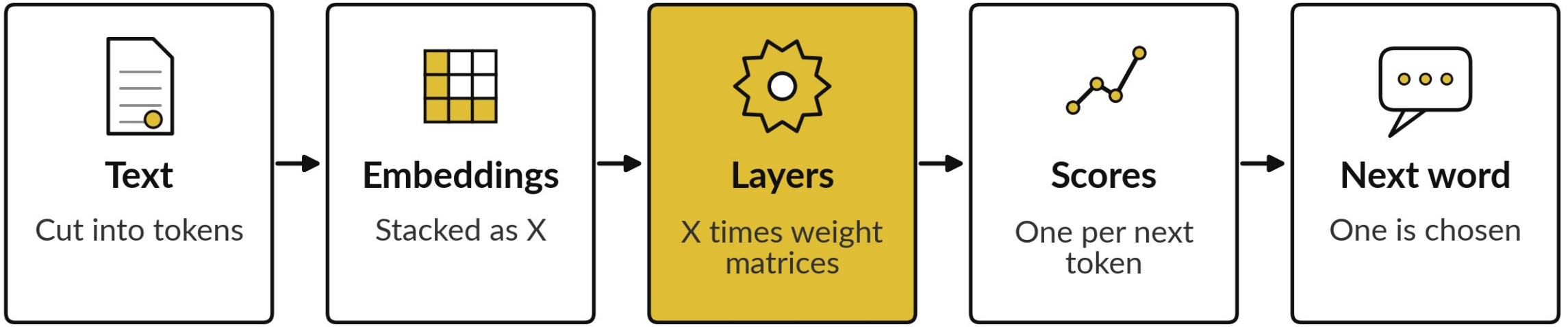
	What you compute	You should get
	Learner table: shape, and entry in row 2, column 3	3 × 3 and 58
	Shape of the transpose of A (2 × 3)	3 × 2
	A x for x = (1, 2, 3)	(14, 32)
	A B, rows 1 and 2; shape of B A	(58, 64), (139, 154); 3 × 3
	Shear moves the corner (1, 3) to	(4, 3)
	Layer 1: y1 = W1 x + b1	(7.5, −1, 3, 4.5)
	Layer 2: W2 y1 + b2	(10.5, 9)
	Parameters: toy network; 1,000 × 1,000 layer	26; 1,001,000

If your number differs, find the step that went wrong. Do not change a number to match.

Four messages you may meet, what they mean and what to do

	What you see	What it means	What to do
	**ValueError: matmul mismatch in the core dimension**	The inner numbers of the shapes differ.	Print A.shape and B.shape. They must match inside.
	**A * B gives an error or the wrong table**	The star multiplies entry by entry.	Use A @ B for the matrix product.
	**#VALUE! or only one cell filled in a spreadsheet**	The formula was not an array over the whole answer range.	Excel: select the gold range, type, press Ctrl Shift Enter. Google Sheets: Enter in the first cell.
	**IndexError for entry 2, 3**	Python counts from 0, so row 2 and column 3 is entry 1, 2.	Subtract 1 from the row and the column number.

Where this appears in AI: most of the arithmetic in each layer is matrix multiplication



Published transformer design (Vaswani et al., 2017): most arithmetic is matrix products.

Where this appears in AI: why graphics processors (GPUs) run models

Why matrix products suit a GPU

Each entry is independent

One dot product per entry of the answer

Work can be shared

A GPU has thousands of small units working at once

Two 4,096 × 4,096 matrices

68,719,476,736 multiplications

Published training reports

Describe clusters of GPUs or similar accelerators

A GPU (graphics processing unit) was first made to draw images, which needs many simple products.

m × n × p multiplications grow fast with size.

Many dot products at once suit a GPU.

Training and running large models need fast matrix products.

Where this appears in AI: check an assistant with your numbers

8 minutes Hands-on activity

Ask an AI assistant to multiply our two matrices.

Give it A with rows (1, 2, 3), (4, 5, 6) and B with rows (7, 8), (9, 10), (11, 12). Ask for A B.

Compare: rows (58, 64) and (139, 154).

Ask for the parameters of a 1,000 by 1,000 layer: expect 1,001,000.

You keep

A note: the assistant answers, your lab numbers, and whether they agree.

Exit check: three questions, then compare with the short answers

	Question	Short answer
	Can a 3 × 4 matrix multiply a 4 × 2 matrix? If yes, what is the shape of the answer?	Yes. The inner numbers are 4 and 4. The answer is 3 × 2.
	What is (1, 2; 3, 4) times the vector (1, 1)? Write the matrix as rows (1, 2) and (3, 4).	Row 1: 1 + 2 = 3. Row 2: 3 + 4 = 7. The answer is (3, 7).
	How many parameters does a layer have with 10 inputs and 5 outputs, with biases?	5 × 10 + 5 = 55.

Save your lab file and add three rules to your sheet

What you keep from today

Keep your saved notebook or workbook, with every check on OK. Keep the idea of the day: a layer is y = W x + b, and the parameters of a model are the numbers in its matrices and biases.

Our question today

A language model has billions of parameters. What are those numbers, and what does the model do with them?

Next session

Session 4: Solving linear systems. We write equations as a matrix, and meet the inverse, the determinant and the rank.

Take away

Your lab file
Saved, all checks OK

The idea
$y = W x + b$

Formula sheet
Three new rules

Next session
Solving linear systems